

Σ X 1 p8 321 Sketch of solutions

a), b), c) $D = -23, -2700, -2700$ is not a square in $\mathbb{Q}, \mathbb{Q}, \mathbb{Q}(\sqrt{2})$

So $G = S_3$. Actually, we must also check $p(x)$ is irreducible.

The hardest case is c) $x^3 - 10$ over $\mathbb{Q}(\sqrt{2})$. It suffices to see that $x^3 - 10$ has no roots in $\mathbb{Q}(\sqrt{2})$. But $x^3 - 10$ is irred over $\mathbb{Q}$ (Eisenstein) so if $\alpha \in \mathbb{Q}(\sqrt{2})$ is a root, then $2 = [\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] \geq [\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$ which is impossible.

To check that -2700 is not a square in $\mathbb{Q}(\sqrt{2})$, it suffices to check that $\sqrt{-3}$ is not in $\mathbb{Q}(\sqrt{2})$. This is obvious as $\mathbb{Q}(\sqrt{2}) \subset \mathbb{R}$

d) e) $\mathbb{Z}/3\mathbb{Z}$ as D is a square

f) $x^4 - 5$ is irr over $\mathbb{Q}$ by Eisenstein's Criterion. The splitting field is $K = \mathbb{Q}(\sqrt[4]{5}, i)$, $[K : \mathbb{Q}] = 8$ is generated by $\sigma: i \rightarrow i, \tau: \sqrt[4]{5} \rightarrow i\sqrt[4]{5}$

so $\sigma^4 = 1, \tau^2 = 1, \tau\sigma\tau = \sigma^3$ so $G(K, \mathbb{Q}) \cong D_4$

$\mathbb{Q}(\sqrt{5}) = K^{\langle \sigma^2, \tau \rangle} \Rightarrow G(K, \mathbb{Q}(\sqrt{5})) \cong \langle \tau^2, \tau \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$

$\mathbb{Q}(i\sqrt{5}) = K^{\langle \sigma^2, \sigma^2 \rangle} \Rightarrow G(K, \mathbb{Q}(i\sqrt{5})) \cong \langle \sigma\tau, \sigma^2 \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$

$\mathbb{Q}(i) = K^{\langle \sigma \rangle} \Rightarrow G(K, \mathbb{Q}(i)) \cong \mathbb{Z}/4\mathbb{Z}$

g) $x^4 - a$ If $a > 0$ this is the same as ex f) so $G = D_4$

If $a < 0$ then $K = \mathbb{Q}(\sqrt[4]{5}w, w^2)$ where $w = \frac{\sqrt{2} + i\sqrt{2}}{2}, w^2 = i, w^4 = -1$

The roots of $x^4 - a$ are $\sqrt[4]{5}w^i, i=1,3,5,7$. We claim that $w^2 \notin \mathbb{Q}(\sqrt[4]{5}w)$ &

so $[K, \mathbb{Q}] = 8$. In fact if $w^2 \in \mathbb{Q}(\sqrt[4]{5}w)$, then $K = \mathbb{Q}(\sqrt[4]{5}w)$ is Galois over $\mathbb{Q}$ with group $\mathbb{Z}/4\mathbb{Z}$. $K \supset \mathbb{Q}(w^2) = \mathbb{Q}(i) \supset \mathbb{Q}$ is the only intermediate field with $[\mathbb{Q}(w^2) : \mathbb{Q}] = 2$. But then $G(K : \mathbb{Q}) \cong \mathbb{Z}/4\mathbb{Z} \cong \langle \sigma \rangle, \sigma: \sqrt[4]{5}w \rightarrow \sqrt[4]{5}w^3$

so $\mathbb{Q}(i) = K^{\langle \sigma^2 \rangle} = \mathbb{Q}(\sqrt[4]{5}w + \sqrt[4]{5}w^5) = \mathbb{Q}(\sqrt[4]{5}\sqrt{2}i) \neq \mathbb{Q}(i)$

so $[K, \mathbb{Q}] = 8$, G is generated by $\tau: \bar{z} \rightarrow \bar{z}, \sigma: \sqrt[4]{5}w \rightarrow \sqrt[4]{5}w^3$

$\tau\sigma\tau = \sigma^3$ so $G \cong D_4$.

h) $\Delta = -27a^2$ is not a square in $\mathbb{Q} \Rightarrow G = S_3$

i) $x^4 + 2$ over $\mathbb{Q}$ is D_4 (see (g)) $\mathbb{Q}(i) = \mathbb{K}^{<G>}$ so

$$G(\mathbb{K}, \mathbb{Q}(i)) \cong \mathbb{Z}/4\mathbb{Z}$$

j) $G \cong (\mathbb{Z}/2\mathbb{Z})^4$. It is enough to show that $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$, $\sqrt{5} \in \mathbb{Q}(\sqrt{2}, \sqrt{3})$

& $\sqrt{7} \in \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$. Now if $\sqrt{3} \in \mathbb{Q}(\sqrt{2})$ then $\sqrt{3} = a + b\sqrt{2}$ $a, b \in \mathbb{Q}$

$$\text{so } 3 = a^2 + 2b^2 + 2ab\sqrt{2} \Rightarrow a \cdot b = 0 \Rightarrow \begin{cases} a=0 \Rightarrow 2b^2=3 \nmid \\ b=0 \Rightarrow a^2=3 \nmid \end{cases}$$

If $\sqrt{3} \in \mathbb{Q}(\sqrt{2}, \sqrt{5})$ then $\sqrt{3} = \alpha + \beta\sqrt{5}$ $\alpha, \beta \in \mathbb{Q}(\sqrt{2}) \Rightarrow 3 = \alpha^2 + 3\beta^2 + 2\alpha\beta\sqrt{5}$

$$\Rightarrow 3 = \alpha^2 + 3\beta^2, 0 = 2\alpha\beta\sqrt{5} \Rightarrow \begin{cases} \alpha=0 \Rightarrow 3 = 3\beta^2 \in \mathbb{Q}(\sqrt{2}) \\ \beta=0 \Rightarrow 3 = \alpha^2 \in \mathbb{Q}(\sqrt{2}) \end{cases}$$

$$3 = \alpha^2 = (a + b\sqrt{2})^2 = a^2 + b^2 \cdot 2 + 2ab\sqrt{2} \Rightarrow \begin{cases} a=0 \rightarrow 3 = 2b^2 \nmid \\ b=0 \rightarrow 3 = a^2 \nmid \end{cases}$$

$$3 = 3\beta^2 = 3(a + b\sqrt{2})^2 = 3(a^2 + 2b^2) + 6ab\sqrt{2} \Rightarrow \begin{cases} a=0 \rightarrow 3 = 6b^2 \Rightarrow \nmid \\ b=0 \rightarrow 3 = 3a^2 \Rightarrow \nmid \end{cases}$$

If $\sqrt{7} \in \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ one also finds a contradiction (See k below)

k) $\mathbb{Z}_2^n$. It suffices to show that $[\mathbb{Q}(\sqrt{p_1} \dots \sqrt{p_n}) : \mathbb{Q}] = 2^n$

We show by induction that $[\mathbb{Q}(\sqrt{p_1/q_1} \dots \sqrt{p_n/q_n}) : \mathbb{Q}] = 2^n$ where

$p_1 \dots p_n, q_1 \dots q_n$ are distinct ^{relatively} primes (possibly $q_i = 1$)

$$n=1: [\mathbb{Q}(\sqrt{p_1/q_1}) : \mathbb{Q}] = 2 \text{ since } \alpha = \sqrt{p_1/q_1}, \alpha^2 = p_1/q_1 \text{ so } \text{irr}(\alpha, \mathbb{Q}, x) = x^2 - p_1/q_1$$

Note $q_1 x^2 - p_1$ is irred by Eisenstein since p_1 is a prime with q_1 .

$$\text{Suppose } K = \mathbb{Q}(\sqrt{p_1/q_1} \dots \sqrt{p_{n-1}/q_{n-1}}) = \mathbb{Q}(\sqrt{p_1/q_1} \dots \sqrt{p_{n-2}/q_{n-2}}) (\sqrt{p_{n-1}/q_{n-1}}) \text{ \& } [K, \mathbb{Q}] = 2^{n-1}$$

Then write $\sqrt{p_n/q_n} = (\alpha + \beta \sqrt{p_{n-1}/q_{n-1}})$ $\alpha, \beta \in \mathbb{Q}(\sqrt{p_1/q_1} \dots \sqrt{p_{n-2}/q_{n-2}})$. Then

$$\frac{p_n}{q_n} = \alpha^2 + \beta^2 \frac{p_{n-1}}{q_{n-1}} + 2\alpha\beta \sqrt{\frac{p_{n-1}}{q_{n-1}}} \Rightarrow \alpha\beta = 0 \text{ If } \beta = 0 \Rightarrow \frac{p_n}{q_n} = \alpha^2 \Rightarrow \sqrt{\frac{p_n}{q_n}} \in \mathbb{Q}(\sqrt{\frac{p_1}{q_1}} \dots \sqrt{\frac{p_{n-2}}{q_{n-2}}})$$

which is impossible by ind. If $\alpha = 0 \Rightarrow \frac{p_n}{q_n} = \beta^2 \frac{p_{n-1}}{q_{n-1}} \Rightarrow \sqrt{\frac{p_n}{q_n}} \in \mathbb{Q}(\sqrt{\frac{p_1}{q_1}} \dots \sqrt{\frac{p_{n-2}}{q_{n-2}}})$

$$\Rightarrow \frac{p_n q_{n-1}}{q_n p_{n-1}} = \beta^2 \Rightarrow \sqrt{\frac{p_n q_{n-1}}{q_n p_{n-1}}} \in \mathbb{Q}(\sqrt{\frac{p_1}{q_1}} \dots \sqrt{\frac{p_{n-2}}{q_{n-2}}}) \text{ which is impossible by ind.}$$

$$l) (x^3-2)(x^3-3)(x^2-2) \quad K = \mathbb{Q}(\sqrt{2}, \sqrt[3]{2}, \sqrt[3]{3}, \sqrt{-3})$$

$\mathbb{Q}(\sqrt[3]{2}, \sqrt{-3}), \mathbb{Q}(\sqrt[3]{3}, \sqrt{-3})$ are Galois over $\mathbb{Q}(\sqrt{-3})$ of deg 3 hence have group $\mathbb{Z}/3\mathbb{Z}$

$$\text{So } G(\mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{3}, \sqrt{-3}), \mathbb{Q}(\sqrt{-3})) \cong \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$$

$$\text{Now } G(\mathbb{Q}(\sqrt{2}, \sqrt{-3}), \mathbb{Q}(\sqrt{-3})) \cong \mathbb{Z}/2\mathbb{Z}, \quad \mathbb{Q}(\sqrt{2}, \sqrt{-3}) \cap \mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{3}, \sqrt{-3}) = \mathbb{Q}(\sqrt{-3})$$

(as ext. degrees are coprime).

$$\text{Therefore } G(K, \mathbb{Q}(\sqrt{-3})) \cong \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$$

$$m) K = \mathbb{C}(\sqrt[n]{t}) \quad G \cong \mathbb{Z}/n\mathbb{Z}$$

$$n) K = \mathbb{R}(i, \sqrt[4]{6}) \quad G(K; \mathbb{R}) \cong D_4$$